

MATÍAS SOLORZA

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Fields of Interest

Macroeconomics, Climate Economics, Finance

Education

University of California, Davis

2022 – Present

PhD in Economics

PhD Candidate, 2025 –

Universidad de Chile

2016

M.A. in Applied Economics (with distinction); Professional Degree in Industrial Engineering (highest distinction)

Research

Working Papers

- Briones, L., and Solorza, M. (2025). *Why Are Recoveries after Natural Disasters So Slow? Evidence from Chile.*

Abstract: We study the macroeconomic consequences of floods and wildfires in Chile using local projections and a unique combination of public and novel administrative data. We document persistent output losses, weak investment despite strong incentives to rebuild productive capital, delayed contractions in commercial credit, and persistent declines in wages following disaster events. While these patterns reveal substantial post-disaster persistence, reduced-form evidence alone cannot identify the distortions responsible for the slow recovery. To address this question, we develop a quantitative accounting framework for post-disaster recoveries based on a dynamic general equilibrium model with capital destruction, financial frictions, productivity shocks, credit supply restrictions, and labor-market wedges. The model is disciplined by the joint dynamics of GDP, investment, credit, wages, and labor-market outcomes, allowing us to quantify the contribution of each distortion to the recovery process. We find that no single distortion explains the persistence of output losses. Instead, productive capital destruction drives the initial contraction, while productivity, financial conditions, credit supply restrictions, and labor-market distortions become quantitatively important at different stages of the recovery. Overall, persistent disaster recoveries in emerging economies arise from the interaction of multiple wedges rather than any single friction. Beyond the evidence for Chile, our framework provides a tractable accounting methodology for quantifying the macroeconomic distortions underlying post-disaster recoveries and evaluating reconstruction policies.

Work in progress

- The Ambiguity Costs of Climate Change(with Frances Moore and Benjamin Collier)
Abstract: Weather risks are fundamentally more difficult to manage under climate change because past experience of weather is potentially unrepresentative of current exposure. The possibility of climate change acts as a global information shock, degrading utility of the historic weather record with implications for actors across the economy making weather-contingent decisions. Here we quantify these ambiguity-related costs for a major climate risk, namely urban flood damages in the United States. Using daily rainfall for 151 US cities from 1970 to 2024, we contrast the 2024 extreme rainfall distribution under a Bayesian model that assumes the rainfall distribution is stationary with one that allows for time-evolving GEV parameters, consistent with the possibility of a changing climate. Estimated tail risks increase notably when allowing the climate to evolve: the 99th percentile rainfall event is larger in 89% of the cities in our sample. Economic costs of these tail risks are amplified by convex damage functions, which we estimate at the city level using flood insurance policy and claim data. Modeling aggregate losses for portfolios at the city, state, regional, and national level, we find geographic diversification only modestly abates these tail risks: the 99th percentile loss event is 4-5 times larger when accounting for evolving climate risk at all levels of aggregation. In a private insurance market these much larger tail risks would substantially increase required capital holdings and the cost of supplying insurance. Our analysis highlights the substantial economic costs of climate change-driven ambiguity in weather extremes and sheds light on recent volatility in US insurance markets.
- How firms' investment decisions change when they are exposed to natural disasters

Publications

- Arroyo Marioli, F., Becerra, J. S., & Solorza, M. (2022). *The credit channel in Chile through the lens of a semi-structural model. Latin American Journal of Central Banking*, 3(2).

Professional Experience

Central Bank of Chile — Medium-Term Forecasting Department

- Senior Economic Analyst, 2021–2022
- Economic Analyst, 2019–2021

Banchile Inversiones — Research Department

Economic Analyst, 2017–2019

Policy Contributions and Notes

- “Impacts of Macroeconomic Policies during the COVID-19 Crisis.” *Monetary Policy Report*, September 2021, Central Bank of Chile (Spanish).
- “Estimation of Structural Parameters for the Chilean Economy.” *Monetary Policy Report*, June 2021, Central Bank of Chile (Spanish).
- “Use of Macroeconomic Models in the Central Bank of Chile.” Central Bank of Chile, December 2020 (Spanish).

(*) Joint work with multiple colleagues at the Central Bank of Chile.

Teaching Assistant Experience (selected)

- Time Series Analysis (PhD), University of California, Davis.
- Macroeconomic Modelling for Emerging Economies (MSc/PhD), Universidad de Chile (2021).
- Introduction to Economics; Macroeconomics; Industrial Organization, Universidad de Chile (2014–2016 and 2022).

Seminars and Conference Presentations

- ASSA Annual Meeting (2027, forthcoming); Annual Meeting of the Chilean Economic Society (SECHI), Chile (2026); Conference on Nature and the Macroeconomy, Chile (2026); Modeling Climate-Economic Dynamics Workshop for Young Scholars, University of Chicago (2026); 1st Piraeus International Economic Conference, Spetses, Greece (2026); World Congress of Environmental and Resource Economists (WCERE), Carcavelos, Portugal (2026); International Conference on Computing in Economics and Finance (2025); Annual Meeting of the Chilean Economic Society (SECHI), Chile (2025); Universidad de Talca, Chile (2025).
- Internal seminars: Central Bank of Chile (2024); UC Davis (2025); NatuRE Policy Lab (2025).
- Pre-PhD: Annual Meeting of the Chilean Economic Society (SECHI), Chile (2016).

Grants, Awards, and Honors

- Dean's Summer Graduate Support Award, University of California, Davis, 2026.
- Selected for the Internship Program for Ph.D. Students, Central Bank of Chile, 2024–2025.
- Travel Award for Conference Participation in Europe, University of California, Davis, 2026.
- Travel Award, “The Macroeconomics of Climate Change” Summer School, CEMFI, 2023.
- PhD Fellowship, Central Bank of Chile, 2022.
- Funding for Master's Thesis, CONICYT (Chile), 2015.
- Scholarship for Master's Program, Center for Applied Economics (CEA), Universidad de Chile, 2015.
- Outstanding Student Award, Universidad de Chile, 2013–2014.

Computational Skills

Matlab; R; Julia; Stata; EViews; $\text{\LaTeX}$; Bloomberg

Languages

English (fluent); Spanish (native)

References

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